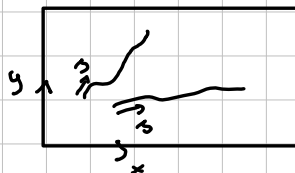


Žitnik: rešene naloge iz M4Fi 2 or sth. (za 1.-3. kol.)

PDE 1. reda

$$a(x, y, u) \frac{\partial u}{\partial x} + b(x, y, u) \frac{\partial u}{\partial y} = \phi(x, y, u)$$



Iščemo krivuljo: karakteristika: $x(s), y(s)$

$$\frac{du(x(s), y(s))}{ds} = \frac{\partial u}{\partial x} \frac{dx}{ds} + \frac{\partial u}{\partial y} \frac{dy}{ds} \rightarrow \text{veržno pravilo}$$

Izberemo:

$$\begin{aligned} \frac{dx}{ds} &= a(x, y, u) \\ \frac{dy}{ds} &= b(x, y, u) \end{aligned} \Rightarrow \frac{du}{ds} = \phi(x, y, u)$$

Burgers: $\frac{\partial s}{\partial t} + s \frac{\partial s}{\partial x} = 0$ ← izpeljali za promet, pogosto tudi pri hidrodinamiki

Začetni pogoj: $s(x, t=0) = \begin{cases} a^2 - x^2 & ; |x| \leq a \\ 0 & ; |x| > a \end{cases}$

$$\frac{dt}{ds} = 1, \quad \frac{dx}{ds} = s, \quad \frac{ds}{ds} = 0 \rightarrow \text{konstante vzdolž karakteristike}$$

$\downarrow$
 $t(s) = s + C \rightarrow$ vseeno
 $t=0 \Leftrightarrow s=0$ ← bolj zberemo

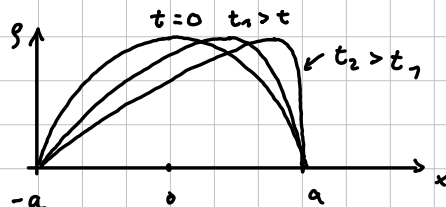
$\hookrightarrow s(s) = \text{konst.}$
 $\hookrightarrow x(s), t(s)$

$$x(s) = \int s(x(s), t(s)) ds = s(x(s), t(s)) \cdot s + D$$

$\downarrow$
 $t=s \Rightarrow x(s) = s(x, s) \cdot s + x_0$
 $\hookrightarrow$ enačba naše karakteristike

$$x(t) = s(x_0, 0)t + x_0$$

← ker je $s = \text{konst.}$, go lahko zberemo ob kateri koli času



kažemo x_0 kot fun. t iz karakt.:

$$x = s(x_0, 0)t + x_0, \quad |x_0| \leq a$$

$$x = (a^2 - x_0^2)t + x_0 = a^2 t - x_0^2 t + x_0 \rightarrow x_0^2 t - x_0 + (x - a^2 t) = 0$$

$$x_0 = \frac{1 \pm \sqrt{1 - 4t(x - a^2 t)}}{2t}$$

$$x=0, t=0 \Rightarrow x_0=0$$

$$x_0 = \lim_{t \rightarrow 0} \frac{1 \pm \sqrt{1 - 4a^2 t^2}}{2t}$$

$$\frac{1+\sqrt{1}}{0} \rightarrow \infty$$

$$\frac{1-1}{0} \rightarrow \frac{0}{0} \rightarrow \text{L'Hospital:}$$

$$x_0 = \lim_{t \rightarrow 0} \frac{-\frac{1}{2}(1 - 4a^2 t^2)^{-1/2} \cdot 8a^2 t}{2} \rightarrow 0$$

$\Rightarrow x_0 = 0 \rightarrow$ to smo želeli $\Rightarrow$ vse je zero, ker je povzroč pravi podznak $\ominus$

naša rešitev, ko se karakteristika ne sekajo

$$s(x, t) = s(x_0(x, t), t=0) = a^2 - \left(\frac{1 - \sqrt{1 - 4t(x - a^2 t)}}{2t} \right)^2$$

če $|x| > a \rightarrow$ karak. se sekajo $\rightarrow$ ne deluje?

← s se seštevja $\rightarrow$ rezanje stolov $\rightarrow$ udarni valovi, naša

metoda odpore $\rightarrow$ da se jih drugače obravnava.

pogosto v hidrodinamiki

$$(x-y) \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = x \quad \text{PDE}$$

$$u(x, y=0) = g(x) \quad \text{zP}$$

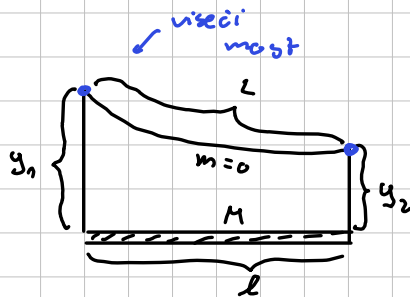
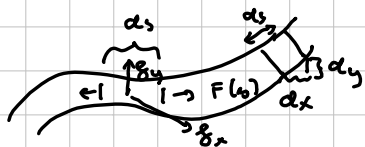
$$\text{Resitev: } u(x, y) = g(1 + (x-y-1)e^{-y}) + y + \frac{1}{2}y^2 + (x-y-1)(1-e^{-y}) \rightarrow DV$$

↓ dolinska gostota i tinit

$$\mu \frac{\partial^2 x}{\partial t^2} = \frac{\partial}{\partial s} \left(F(s) \frac{\partial x}{\partial s} \right) + g_x$$

$$\mu \frac{\partial^2 y}{\partial t^2} = \frac{\partial}{\partial s} \left(F(s) \frac{\partial y}{\partial s} \right) + g_y$$

$$\mu = \frac{dm}{ds}, \quad g_x = \frac{dF_{\sin \alpha}}{ds}$$



R.P.

$$\frac{dy}{dt} = 0$$

$$\frac{dx}{dt} = 0$$

napetost strune
veliki x smeri

Kvazistatist · $ds^2 = dx^2 + dy^2$

x smer: $\frac{\partial}{\partial s} \left(F(s) \frac{\partial x}{\partial s} \right) = -g_x = 0 \Rightarrow F(s) \frac{\partial x}{\partial s} = \text{konst.} = F_0$

y smer: $\frac{\partial}{\partial s} \left(F(s) \frac{\partial y}{\partial s} \right) = -g_y$

$$dF_{201} = \frac{Mg dx}{l} \Rightarrow g_y = -\frac{Mg}{l} \frac{dx}{ds}$$

↳ ker nimamo

unicke stezo ampak most stezo

$$\frac{\partial}{\partial s} \left(F(s) \frac{\partial y}{\partial s} \right) = \frac{Mg}{l} \frac{dx}{ds}$$

$$\frac{\partial}{\partial s} \left(\frac{F_0}{\frac{\partial x}{\partial s}} \cdot \frac{\partial y}{\partial s} \right) = \frac{Mg}{l} \frac{dx}{ds}$$

$$\frac{\partial}{\partial s} = \frac{\partial x}{\partial s} \frac{\partial}{\partial x}$$

$$\frac{\partial}{\partial s} \left(\frac{\partial y}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial y}{\partial x} \right) \frac{\partial x}{\partial s} = \frac{\partial^2 y}{\partial x^2} \frac{\partial x}{\partial s}$$

$$\frac{\partial}{\partial s} \left(F_0 \cdot \frac{\partial y}{\partial x} \right) = \frac{Mg}{l} \frac{dx}{ds}$$

$$F_0 \frac{\partial x}{\partial s} \frac{\partial^2 y}{\partial x^2} = \frac{Mg}{l} \frac{dx}{ds} \rightarrow \frac{\partial^2 y}{\partial x^2} = \frac{Mg}{l F_0} \rightarrow \frac{\partial y}{\partial x} = \frac{Mg}{l F_0} x + C_1$$

$$y(x) = \frac{Mg}{l F_0} \frac{x^2}{2} + C_1 x + C_2$$

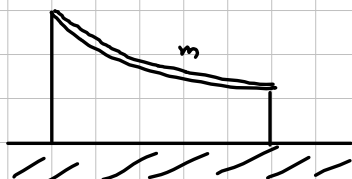
i) $y(0) = \frac{Mg}{l F_0} \frac{0^2}{2} + C_1 \cdot 0 + C_2 \rightarrow y_1 = C_2$

ii) $y(l) = \frac{Mg}{l F_0} \frac{l^2}{2} + C_1 l + y_1 \rightarrow C_1 = \frac{1}{l} \left(y_2 - y_1 - \frac{Mg l}{2 F_0} \right)$

iii) $L = \int ds = \int \sqrt{dx^2 + dy^2} = \int \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx$

$$F_0 = \dots$$

Če imamo struno o teži:

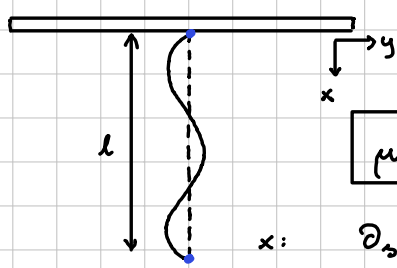


dolinska
gostota
 $g_y = -\mu g$
↳ tež se ne spreminja
več po x, ampak
po dolžini i struno

Catenary

$$\frac{\partial}{\partial s} \left(F_0 \frac{dy}{dx} \right) = \mu g \rightarrow F_0 \frac{dy}{dx} = \mu g s + C \Rightarrow \dots \Rightarrow$$

$$y = a \cosh\left(\frac{x}{a}\right), \quad a = \frac{F_0}{\mu g}$$



Majori odčitki v y:

$$ds^2 = dx^2 + dy^2 \approx dx^2 \Rightarrow x = s$$

$$\mu \frac{\partial^2 x}{\partial t^2} = \frac{\partial}{\partial s} \left(F(s) \frac{\partial x}{\partial s} \right) + f_x$$

$$x: \quad \frac{\partial}{\partial s} F(s) = -f_x = \frac{dF_{\text{um},x}}{ds} \stackrel{\mu g}{\rightarrow} F_{\text{um},x} = mg$$

$$dF_{\text{um},x} = \mu dx g$$

$$F(x) = \mu g x + F_0$$

$$mg = F(0) = F_0 \rightarrow F(x) = \mu g (l - x)$$

$$y: \quad \mu \frac{\partial^2 y}{\partial t^2} = \frac{\partial}{\partial x} \left(F(x) \frac{\partial y}{\partial x} \right) + f_y$$

$$\mu \frac{\partial^2 y}{\partial t^2} = \mu g \left(-\frac{\partial y}{\partial x} + (l-x) \frac{\partial^2 y}{\partial x^2} \right)$$

$$\frac{\partial^2 y}{\partial t^2} = g \left[(l-x) \frac{\partial^2 y}{\partial x^2} - \frac{\partial y}{\partial x} \right]$$

$$y(x, t) = X(x) \cdot T(t)$$

$$X\ddot{T} = g(l-x)X''T - gX'T \quad \text{dani imaginarno}$$

$$\frac{\ddot{T}}{T} = g(l-x) \frac{X''}{X} - g \frac{X'}{X} \stackrel{\downarrow}{=} -\omega^2$$

$$t: \quad \ddot{T} + \omega^2 T = 0 \rightarrow T(t) = A \cos(\omega t) + B \sin(\omega t)$$

$$x: \quad g(l-x)X'' - gX' + \omega^2 X = 0 \rightarrow \text{to lahko predelamo na Bessla}$$

Bessel: $z^2 y'' + z y' + (z^2 - n^2) y = 0 \rightarrow y = A J_n(z) + B Y_n(z)$
 $z = \sqrt{l-x}$
 $\frac{dX}{dx} = \frac{dX}{dz} \frac{dz}{dx} = \frac{dX}{dz} \cdot \frac{-1}{2\sqrt{l-x}} = -\frac{dX}{dz} \frac{1}{2z}$
 $\frac{d^2 X}{dx^2} = \frac{d}{dx} \left(-\frac{dX}{dz} \frac{1}{2z} \right) = + \frac{1}{2z} \frac{d}{dz} \left(\frac{1}{2z} \frac{dX}{dz} \right) = \frac{1}{2z} \left(-\frac{1}{2z^2} \frac{dX}{dz} + \frac{1}{2z} \frac{d^2 X}{dz^2} \right) \rightsquigarrow \text{nesemo v našo enačbo } \underline{x}:$

$$z^2 \frac{1}{2z} \left(\frac{1}{2z} \frac{d^2 X}{dz^2} - \frac{1}{2z^2} \frac{dX}{dz} \right) + \frac{1}{2z} \frac{dX}{dz} + \frac{\omega^2}{g} X = 0 \quad | \cdot 4z^2$$

$$z^2 \frac{d^2 X}{dz^2} - z \frac{dX}{dz} + 2z \frac{dX}{dz} + 4z^2 \frac{\omega^2}{g} X = 0$$

$$z^2 \frac{d^2 X}{dz^2} + z \frac{dX}{dz} + 4z^2 \frac{\omega^2}{g} X = 0$$

$$\tilde{z} = 2z \sqrt{\frac{\omega}{g}} = 2\omega \sqrt{\frac{l-x}{g}}$$

$$\tilde{z}^2 \frac{d^2 X}{d\tilde{z}^2} + \tilde{z} \frac{dX}{d\tilde{z}} + \tilde{z}^2 X = 0 \rightarrow \text{Bessel}$$

$$X(x) = C J_0 \left(2\omega \sqrt{\frac{l-x}{g}} \right) + D Y_0 \left(2\omega \sqrt{\frac{l-x}{g}} \right)$$

Robni pogoji:

$$i) \quad y(x=0, t) = 0 \rightarrow \text{pritjena struna}$$

$$ii) \quad y(x, t) < \infty \rightarrow y \text{ nikoli ne divergira} \rightarrow \text{uporaben RP}$$

$$\Rightarrow \forall Y_n(0) \rightarrow -\infty \quad \text{nujno } D=0, \text{ ker ne smemo imeti } Y_0 \text{ (} Y_0(0) \text{ pri } x=l \text{)}$$

$$\Rightarrow X(0) = 0 \rightsquigarrow J_0 \left(2\omega \sqrt{\frac{l-x}{g}} \right) \Big|_{x=0} = 0$$

$$J_0 \left(2\omega \sqrt{\frac{l}{g}} \right) = 0 \rightarrow \text{nekončno rešitev}$$

$$\Downarrow \quad 2\omega \sqrt{\frac{l}{g}} = \xi_n^{(0)} \rightarrow \text{n-ta ničla ničte Besslove fun.}, n = 1, 2, \dots$$

$$\omega_n = \frac{1}{2} \sqrt{\frac{g}{l}} \xi_n^{(0)}$$

$\rightarrow$ lastne frekv. $\rightarrow$ tudi če jo drugače def., dobimo končni rezultat enak

$$y(x, t) = \sum_{n=1}^{\infty} (A_n \cos \omega_n t + B_n \sin \omega_n t) J_0 \left(2\omega_n \sqrt{\frac{l-x}{g}} \right)$$

$\rightarrow$ če en člen reši enačbo, tudi vsota reši enačbo
 $\rightarrow$ ni vedno vsota rešitev

$\rightarrow$ potrebujemo homogen PDE in homogene RP

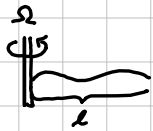
začetni pogoji:

$$y(x, t=0) = \sum_{n=1}^{\infty} A_n J_0 \left(2\omega_n \sqrt{\frac{l-x}{g}} \right)$$

$\rightarrow$ orton. (podobno kot baza)

Podobna naloga za doma:

Vrteča se struna $\rightarrow$ zanimajo nas majhna nihanja



Legendrove enačbe

$$\omega_n = \Omega \sqrt{\frac{n(n+1)}{2}}$$

$$\text{Rezultat: } y(x, t) = \sum_{n=1}^{\infty} P_n\left(\frac{x}{l}\right) [A_n \sin(\omega_n t) + B_n \cos(\omega_n t)]$$

Struno pripramo s konst. silo napetanja:

$$F = \text{konst.}$$

$$\Rightarrow c^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}, \quad c^2 = \frac{F}{\mu}$$

d'Alembertov princip: $u(x, t) = g(x - ct) + g(x + ct)$

2 karakteristiki:

- $x - ct = \text{konst.}$
- $x + ct = \text{konst.}$

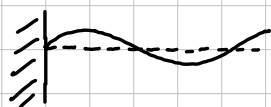
d'Alembertova rešitev: $x \in \mathbb{R}$

oblika strune ob času nič

2P: $u(x, 0)$ in $\frac{\partial u}{\partial t}(x, 0)$

hitrost na zač. (ali jo samo govorimo ali...)

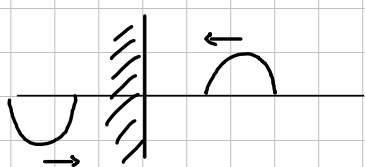
$$u(x, t) = \frac{1}{2} [u(x - ct, 0) + u(x + ct, 0)] + \frac{1}{2c} \int_{x-ct}^{x+ct} \frac{\partial u}{\partial t}(\xi, 0) d\xi$$



$$\text{PDE: } c^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$$

2P: $u(x, 0)$ in $\frac{\partial u}{\partial t}(x, 0)$

RP: a) trdno vpetje
b) prosto vpetje



Če bi uporabili samo d'Alemberta, bi valji, ko bi prišel do leve zgradbe $\rightarrow$ z - preverali o val na drugo stran strune

RP: a) $u(x=0, t) = 0$

$\tilde{u}(x, t), x \in \mathbb{R}$

2P: i) $\tilde{u}(x, 0) = \begin{cases} u(x, 0); & x > 0 \\ 0 & ; \text{ sicer} \end{cases}$

ii) $\frac{\partial \tilde{u}}{\partial t}(x, 0) = \begin{cases} \frac{\partial u}{\partial t}(x, 0); & x > 0 \\ 0 & ; \text{ sicer} \end{cases}$

$$\tilde{u}(x, t) = \frac{1}{2} [\tilde{u}(x - ct, 0) + \tilde{u}(x + ct, 0)] + \frac{1}{2c} \int_{x-ct}^{x+ct} \frac{\partial \tilde{u}}{\partial t}(\xi, 0) d\xi$$

$$u(x, t) = \tilde{u}(x, t) - \tilde{u}(-x, t)$$

$$\text{ali } u \text{ ustreza PDE: } \frac{\partial^2 u}{\partial x^2}(x, t) = \frac{\partial^2 \tilde{u}(x, t)}{\partial x^2} - \frac{\partial^2 \tilde{u}(-x, t)}{\partial x^2} \cdot (-1)^2$$

$$= \frac{1}{c^2} \frac{\partial^2 \tilde{u}}{\partial t^2}(x, t) - \frac{1}{c^2} \frac{\partial^2 \tilde{u}}{\partial t^2}(-x, t) = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}(x, t) \quad \checkmark$$

2P: i) $u(x, 0) = \tilde{u}(x, 0) - \tilde{u}(-x, 0) = u(x, 0)$

ii) $\frac{\partial u}{\partial t}(x, 0) = \frac{\partial \tilde{u}}{\partial t}(x, 0) - \frac{\partial \tilde{u}}{\partial t}(-x, 0) \quad \checkmark$

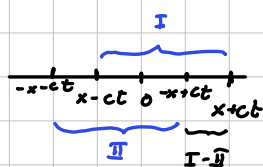
RP: $u(0, t) = \tilde{u}(0, t) - \tilde{u}(-0, t) = 0$

$$u(x, t) = \frac{1}{2} [\tilde{u}(x - ct, 0) + \tilde{u}(x + ct, 0)] + \frac{1}{2c} \int_{x-ct}^{x+ct} \frac{\partial \tilde{u}}{\partial t}(\xi, 0) d\xi -$$

$$- \frac{1}{2} [\tilde{u}(-x - ct, 0) + \tilde{u}(-x + ct, 0)] - \frac{1}{2c} \int_{-x-ct}^{-x+ct} \frac{\partial \tilde{u}}{\partial t}(\xi, 0) d\xi$$

$$x > ct: u(x, t) = \frac{1}{2} [u(x - ct, 0) + u(x + ct, 0)] + \frac{1}{2c} \int_{x-ct}^{x+ct} \frac{\partial u}{\partial t}(\xi, 0) d\xi$$

$$0 < x < ct: u(x, t) = \frac{1}{2} [u(x + ct, 0) - u(-x + ct, 0)] + \frac{1}{2c} \int_{-x+ct}^{x+ct} \frac{\partial u}{\partial t}(\xi, 0) d\xi$$

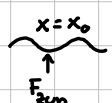


b) prosto vpetje:

$$F_{y, \text{zun}} = 0$$



$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} + \frac{g_y}{\mu}$$



$$g_y = F_{\text{zun}} \cdot \delta(x - x_0)$$

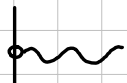
$$\int g_y dx = F_{\text{zun}} \int \delta(x - x_0) dx = F_{\text{zun}}$$

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} + \frac{F_z}{\mu} \delta(x - x_0) \quad \Bigg| \int_{x_0-\epsilon}^{x_0+\epsilon} \dots dx, \quad \epsilon > 0$$

dobiti moramo zvezno strun, ker imamo konst. dielektrično silo $\rightarrow$ pospešila ne morejo kar skakati

$$0 = c^2 \frac{\partial u}{\partial x} \Bigg|_{x_0-\epsilon}^{x_0+\epsilon} + \frac{F_z}{\mu}, \quad c^2 = \frac{F}{\mu}$$

$$\Rightarrow \boxed{F_z = -F \frac{\partial u}{\partial x} \Bigg|_{x_0-\epsilon}^{x_0+\epsilon}}$$

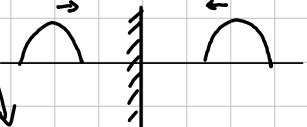


$$F_z = -F \frac{\partial u}{\partial x}(x=0, t)$$

$$\text{RP: } F_z = 0 = -F \frac{\partial u}{\partial x}(x=0, t) \Rightarrow \boxed{\frac{\partial u}{\partial x}(x=0, t) = 0}$$

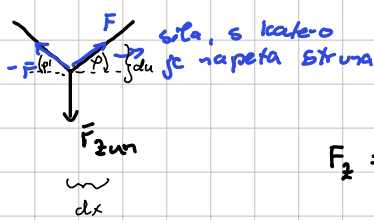
2 RP: Dirichletov RP: $u(x=0, t) = 0$

Neumannov RP: $\frac{\partial u}{\partial x}(x=0, t) = 0$



$$u(x, t) = \tilde{u}(x, t) + \tilde{u}(-x, t)$$

Fizikalna predstav:



$$\tan \varphi = \frac{du}{dx}$$

majhni koti
 $\sin \varphi \approx \tan \varphi$

$$F_z = F \sin \varphi - F \sin \varphi' = F \frac{du}{dx} =$$

$$= F \frac{du}{dx} \Bigg|_{x_0+} - F \frac{du}{dx} \Bigg|_{x_0-}$$